

Alberta opioid deaths around COVID-19

An interrupted time series with placebo and NB cross-checks

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github.com/Adilbek123/alberta-substance-use-trends

What I asked, and how I tried to answer it

Question

Did Alberta's opioid death rate shift at a level higher than the pre-COVID trend can explain, starting from 2020 Q2 (first full quarter under the public health emergency)?

Data

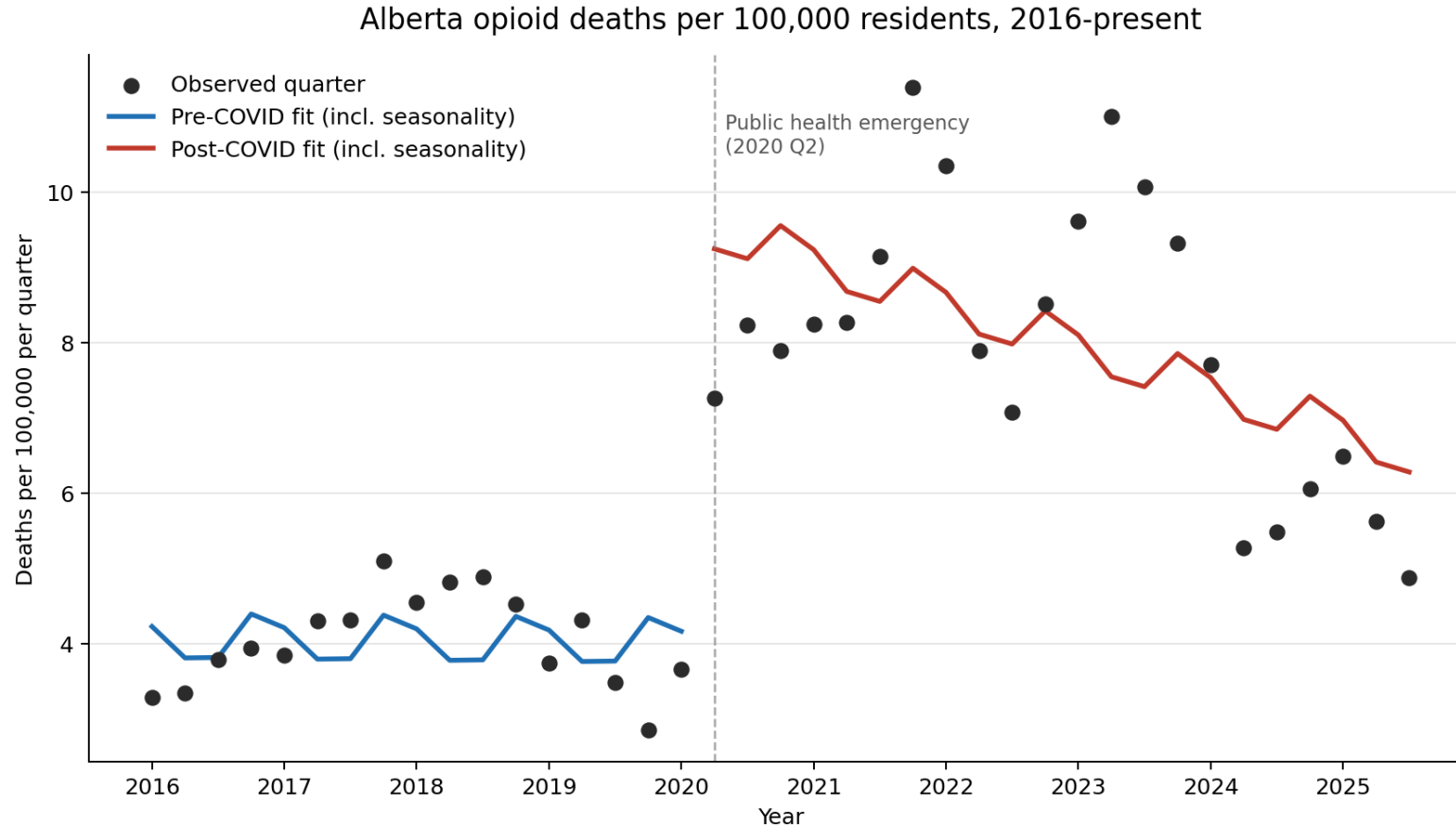
PHAC Health Infobase substance-related harms data + StatCan Table 17-10-0009-01. 39 quarters of Alberta opioid deaths, 2016 Q1 to 2025 Q3, normalised to deaths per 100k residents.

Method

Interrupted time series with a level and slope change around 2020 Q2. Quarter fixed effects for seasonality. Newey-West HAC SE with small-sample correction.

Robustness: HAC lag sensitivity, donut (drop 2020 Q1), seven placebo cutoffs in the pre-period, negative binomial cross-check with log-population offset.

Sharp, large, robust break at 2020 Q2



Sources: PHAC Health Infobase substance-related harms data; StatCan Table 17-10-0009-01.

Level shift at 2020 Q2 (segmented regression with quarter FE, HAC SE 3-lag, small-sample correction): +5.50 per 100k [95% CI +3.60, +7.39].

Main spec

Level shift at 2020 Q2:

+5.50 per 100k per quarter

(95% CI +3.60, +7.39, $p < 0.001$)

At mean AB pop: ~+246 deaths/quarter

Robustness

HAC L=1..6: +5.50 (SE 0.79–0.98)

Donut (drop 2020 Q1): +5.34

NB rate ratio: 2.43 (1.80, 3.26)

Placebo cutoffs

Seven fake cutoffs in pre-period:

All point opposite direction.

Largest |placebo|: 1.86

Real jump (+5.50) is ~3× larger.

What this identifies — and what it doesn't

What the estimate captures

The total effect of the COVID-19 pandemic on Alberta opioid mortality.

Including the bundle of policy and social responses the pandemic produced: supply-chain disruption in the unregulated drug supply, reduced harm-reduction service capacity, mental-health service disruption, social isolation, economic shock. These are downstream of COVID, not parallel causes.

What it does not do

Decompose that total into its component channels.

The estimate cannot tell you what share came from supply toxicity vs. service capacity vs. isolation vs. economic shock. A peer-province comparison would not fix this — all provinces faced COVID and its policy response at essentially the same time, so the comparison identifies Alberta-specific deviation, not a clean COVID-only channel.

Realistic next steps

1. Mechanism decomposition: triangulate supply-side, service-capacity, and outcome signals to attribute share
2. Heterogeneity: by age, sex, zone, substance type — where the level shift concentrated
3. Post-period dynamics: explicitly model the rise-and-decline shape, not just the immediate level shift
4. Cumulative excess deaths through end-of-sample as a policy-relevant quantity